

Killian Gerard

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PROFILE

Mechatronics BEng graduate (2:1) with two engineering internships in live energy operations: Vitol Group on UK gas turbines and VIVO Energy (Shell licensee) on LPG and oil infrastructure in Morocco. Hands-on experience building machine-learning models in Python (battery-life prediction and high-speed-rail vibration analysis) and end-to-end CAPEX/OPEX modelling and oil-demand forecasting from fieldwork. Dual French & British national and trilingual (French, English, Spanish). Completed IE Madrid's Summer School in Finance Fundamentals in summer 2026, ahead of joining the MSc Fintech programme in September 2026.

PROFESSIONAL EXPERIENCE

JUL–AUG 2025 **Engineering Internship**

VIVO Energy (Shell Licensee) · Morocco

- Shadowed engineers across 3+ LPG and oil-infrastructure disciplines, supporting project development and management.
- Learned end-to-end CAPEX/OPEX modelling, covering Excel structure, oil-demand forecasting and tank-reserve analytics with fail-safe systems.
- Monitored oil-tank construction and welding under strict safety regulation; redesigned the division's engineer-onboarding deck.

JUL–AUG 2022 **Engineering Shadowing**

VPI Damhead Creek, Vitol Group · UK

- Shadowed real-time performance of Mitsubishi 701F3 gas turbines (~800 MW combined).
- Adjusted operational output against regional daily demand; supported procurement and formal reporting.

KEY PROJECTS

- 3RD YR ML for Remaining Useful Life of Li-ion batteries. LSTM on the Stanford/MIT dataset, MAPE < 0.07%.
- 3RD YR Vibration analysis: high-speed-rail suspension concept with ML anomaly detection.
- 2ND YR Line-following robot firing ping-pong balls at a target after a set distance.
- 1ST YR IMechE robot car: drives into a wall and reverses precisely to its start position.
- 1ST YR Scalable miniature wind-turbine aerofoil, built with JavaFoil, SolidWorks and wind-tunnel testing.

EDUCATION

IE University, Madrid

MSc Fintech, from Sept 2026

Summer School — Finance Fundamentals, Jun–Jul 2026

Lancaster University

Mechatronics Engineering BEng (Hons), 2:1, 2023–2026

University of Nottingham

Physics w/ Astronomy & Foundation, 2022–2023

Colfe's School, London

Private secondary & sixth form, 2015–2021

LANGUAGES

French & English (native), Spanish (proficient),
Arabic / Portuguese / Italian (elementary).

LEADERSHIP & AWARDS

EPALS Vice-President · RAF (CCF) Cadet, Corporal · Duke of Edinburgh's Award, Silver & Bronze.

TECHNICAL SKILLS

Python (Pandas, NumPy, TensorFlow/Keras, scikit-learn, Matplotlib, Gradio), MATLAB, C/C++ embedded. Machine learning, time-series & predictive modelling. SolidWorks, AutoCAD, JavaFoil. CAPEX/OPEX modelling.

CERTIFICATIONS

A-Levels: Maths (B), Physics (B), Spanish (A). GCSEs: 9 subjects (2 A*, 1 A and grades 8–5). Foundation Year passed with Merit. First Flight Certificate, Piper Warrior PA-28 (2019). Kaggle certificates in Python, machine-learning and data-science fundamentals. Piano (Grade 5) and violin (Grade 4). Corporal, RAF (CCF) Cadets.